

Dominant Beta-Thalassemia (Inclusion-Body β -Thalassemia / Hyperunstable Hemoglobinopathy)

Disease: Dominant Beta-Thalassemia **MONDO ID:** MONDO:0011381 **Category:** Genetic (autosomal-dominant hemoglobinopathy) **Report type:** Comprehensive disease-characterization report for knowledge-base population

Summary

Dominant β -thalassemia is a rare autosomal-dominant hemoglobinopathy in which a **single mutant *HBB* (β -globin) allele** is sufficient to produce clinically overt disease in a heterozygote — a fundamental contrast with the far more common recessive β -thalassemias, in which heterozygotes are asymptomatic carriers. The disease is caused predominantly by **exon-3 missense, nonsense (nonsense-mediated-decay-escaping), or frameshift/elongating mutations** in *HBB* that yield a *translated* but **hyperunstable β -globin chain**. This aberrant chain retains the ability to bind heme, precipitates as **protease-resistant inclusion (Heinz) bodies** in erythroid precursors and mature red cells, and behaves as a **toxic, dominant-negative gene product** rather than a simple loss-of-function allele ([PMID: 1971109](#), [PMID: 34957901](#)).

Mechanistically, the disease is best understood as a **proteostasis / protein-aggregation disorder of the erythron**. The unstable β -globin (together with the resulting excess of unpaired α -globin) overwhelms and escapes the erythroid **ubiquitin-proteasome and autophagy quality-control machinery**, precipitates with attached heme, generates **reactive oxygen species**, and drives **apoptosis of erythroblasts (ineffective erythropoiesis)** and **splenic destruction of inclusion-laden red cells (hemolysis)**. The net clinical picture is a **thalassemia-intermedia-like phenotype**: moderate hemolytic anemia, reticulocytosis, Heinz bodies, splenomegaly, and secondary iron overload. Severity is modified by co-inherited genetic factors — notably **co-inheritance of α -thalassemia** (which reduces the α/β chain imbalance and softens the phenotype) and **HbF-boosting variants** ([PMID: 24432801](#), [PMID: 28651846](#)).

Because affected heterozygotes are symptomatic, the dominant form is **not enriched by malaria selection** and therefore occurs **pan-ethnically as isolated families or de novo cases** (only ~12 families were known worldwide with a third-exon nonsense allele at the time of one foundational survey) — unlike recessive β -thalassemia, which is concentrated in historically malaria-endemic regions ([PMID: 8184583](#)). Management follows the broader β -thalassemia framework: **supportive care** (transfusion, iron chelation, and often **splenectomy**, which can restore transfusion independence), **disease-modifying agents** (luspatercept, mitapivat), and potentially **curative hematopoietic stem-cell transplantation or gene addition/editing therapy** (betibeglogene autotemcel; exagamglogene autotemcel) ([PMID: 42584024](#), [PMID: 142356](#)).

1. Disease Information

Overview. Dominant β -thalassemia is a form of β -thalassemia in which heterozygosity for a single *HBB* mutation produces a clinically significant hemolytic/dyserythropoietic anemia. It is also known as **inclusion-body β -thalassemia**, **dominantly inherited β -thalassemia**, and **hyperunstable hemoglobinopathy (HUH)** — the latter term explicitly equated with "dominantly inherited β -thalassemia" in the literature and classified as "a relatively rare form of congenital hemolytic anemia" ([PMID: 24432801](#)). The defining feature is that the mutant allele encodes a *translated*, structurally abnormal β -globin that forms inclusion bodies, so a single copy causes disease.

Key identifiers.

Resource	Identifier
MONDO	MONDO:0011381
OMIM (β -thalassemia; <i>HBB</i>)	#603902 (beta-thalassemia); <i>HBB</i> gene 141900
Gene	<i>HBB</i> , HGNC:4827, NCBI Gene 3043, chromosome 11p15.4
MeSH	beta-Thalassemia (D017086)
ICD-10	D56.1 (Beta thalassemia)
ICD-11	3A50.1 (Beta thalassaemia)

Synonyms / alternative names: dominant beta-thalassemia; dominantly inherited β -thalassemia; inclusion-body β -thalassemia; hyperunstable hemoglobinopathy (HUH); (historically overlapping with) congenital Heinz-body hemolytic anemia due to unstable β -globin variants.

Nature of evidence. The knowledge base for this entity is derived from **aggregated disease-level resources and case/family reports** (individual probands and pedigrees with defined *HBB* variants), plus mechanistic work in **in-vitro** systems and **mouse models**. It is not primarily an EHR/population-cohort disease because of its rarity.

2. Etiology

Primary cause (genetic). Heterozygous mutations in the β -globin gene *HBB* on chromosome 11p15.4. Unlike recessive β -thalassemia (where a single defective allele is silent), the dominant form arises from **specific mutation classes — predominantly in exon 3 — that produce an abnormal but translated globin product** (PMID: 1971109). Representative causal variants documented in the literature:

Variant (common name / HGVS)	Type	Reference
Hb Hradec Kralove, β 115(G17)Ala→Asp	Missense (exon 3)	PMID: 7693620
Codon 121 (G→T), codon 112 (T→A)	Nonsense (exon 3, NMD-escaping)	PMID: 8184583
Hb Dieppe, β 127(H5)Gln→Arg	Missense (exon 3)	PMID: 34957901
HBB:c.313delA	Frameshift / elongating	PMID: 34271589
Hb Grand Junction, HBB:c.348_349delinsG; p.His117IlefsX42	Frameshift (codons 115/116)	PMID: 24432801

Genetic risk factors. The causal variant itself is the sole necessary risk factor. **Modifier genes** influence severity: the **α -globin genotype** (co-inherited α -thalassemia is protective; see below) and **HbF-modifying loci** (e.g., *BCL11A*, *HBS1L-MYB*) provide "some prediction of disease severity for β thalassemia" (PMID: 28651846). **AHSP (α -hemoglobin-stabilizing protein)** is a candidate modulatory factor via its role in escorting free α -globin (PMID: 31894534).

Environmental risk factors. None are established as causal. Oxidative stressors could theoretically aggravate hemolysis in unstable-hemoglobin states, but the disease is fundamentally monogenic. Family history (an affected parent) is the main non-modifiable factor; **de novo mutation** is well documented.

Protective factors. The best-established **genetic protective factor is co-inherited α -thalassemia** (e.g., a -3.7 kb single α -gene deletion), which "leads to a decreased imbalance between α and β chain formation, and subsequently a milder phenotype" (PMID: 24432801). **Elevated HbF** (high-HbF genotypes/HbF-inducing modifiers) is likewise ameliorating.

Gene–environment interactions. Not a significant feature. Disease expression is governed largely by the primary lesion plus genetic modifiers (α -genotype, HbF).

3. Phenotypes

The phenotype resembles **β -thalassemia intermedia** / congenital non-spherocytic hemolytic anemia. In Hb Hradec Kralove heterozygotes the picture was "moderate anemia, reticulocytosis, nucleated red cells, target cells, and other red cell changes, Heinz body formation, and splenomegaly," with marked compensatory increase in fetal-hemoglobin synthesis (PMID: 7693620). Czech/Slovak dominant alleles presented as thalassemia intermedia with Heinz bodies in peripheral erythrocytes, and — importantly — **severity varied markedly even within families** carrying identical mutations (PMID: 8184583).

Phenotype	Type	HPO suggestion	Onset / severity / frequency
Hemolytic anemia	Lab / clinical	HP:0001878 (Hemolytic anemia); HP:0001903 (Anemia)	Childhood onset; moderate; near-universal
Reticulocytosis	Lab abnormality	HP:0001923 (Reticulocytosis)	Chronic compensatory; frequent
Heinz bodies / red-cell inclusions	Lab / morphologic	(Heinz body inclusion; verify HPO ID)	Hallmark; frequent
Splenomegaly	Clinical sign	HP:0001744 (Splenomegaly)	Progressive; common
Jaundice / hyperbilirubinemia	Clinical / lab	HP:0000952 (Jaundice); HP:0002904 (Hyperbilirubinemia)	Chronic; common
Microcytic hypochromic red cells	Lab	HP:0001935 (Microcytic anemia)	Frequent
Nucleated RBCs / abnormal morphology	Lab	HP:0012132 (Erythroid abnormality)	Frequent
Elevated HbA2 / HbF	Lab	(elevated HbF; verify HPO ID)	Frequent
Iron overload (secondary)	Lab / clinical	HP:0011031 (Abnormal iron homeostasis)	Late; treatment/disease related

Quality-of-life impact. Chronic anemia, fatigue, splenomegaly, and (when present) transfusion dependence and iron-overload complications impair daily functioning. QoL burden parallels that documented for β -thalassemia intermedia/transfusion-dependent thalassemia; generic tools (SF-36, EQ-5D) are used in the broader thalassemia literature. Disease-specific QoL data for the dominant subtype specifically were **not identified**.

4. Genetic / Molecular Information

Causal gene. *HBB* (β -globin; HGNC:4827; NCBI Gene 3043; OMIM 141900), chromosome 11p15.4.

Pathogenic variants. Dominant β -thalassemia is characteristically caused by: - **Missense** mutations in **exon 3** producing hyperunstable globins (e.g., Hb Hradec Kralove $\beta 115\text{Ala}\rightarrow\text{Asp}$; Hb Dieppe $\beta 127\text{Gln}\rightarrow\text{Arg}$) — "certain missense mutations in exon 3, however, produce unstable globins causing a dominant β -thal phenotype or hemolytic anemia in heterozygotes" (PMID: 34957901). - **Nonsense** mutations in exon 3 that **escape nonsense-mediated decay** (e.g., codon 121 G $\rightarrow$ T, codon 112 T $\rightarrow$ A), so a truncated toxic chain is translated (PMID: 8184583). - **Frameshift/elongating** variants (e.g., HBB:c.313delA producing a β -chain elongated by 10 residues; Hb Grand Junction p.His117IlefsX42) (PMID: 34271589, PMID: 24432801).

Variant classification. These variants are **pathogenic (dominant)** by family segregation and functional data. **Origin is germline** (inherited or de novo); this is not a somatic/oncologic disease. **Allele frequencies** are effectively absent from population databases (private/rare family-specific alleles), consistent with the lack of malaria-driven selection.

Functional consequences. The defining consequence is a **toxic gain-of-function / dominant-negative** effect: the abnormal chain "binds heme and produces aggregations that are relatively resistant to proteolytic degradation" (PMID: 1971109). For frameshift/elongating alleles, the phenotype is "mainly related to the stability of mutant mRNA, the degradation of mutant proteins" — a combination of mRNA-stability and protein-degradation determinants rather than simple quantitative deficiency (PMID: 34271589).

Modifier genes: α -globin genotype (co-inherited α -thalassemia), HbF loci (*BCL11A*, *HBS1L-MYB*), and *AHSP*.

Epigenetic information / chromosomal abnormalities. No specific epigenetic signature or large-scale chromosomal abnormality is characteristic; the disease is a point-mutation/small-indel disorder of *HBB*. (HbF induction pharmacology engages γ -globin regulation but is therapeutic, not etiologic.)

5. Environmental Information

Dominant β -thalassemia is a **monogenic disease with no established environmental, toxic, occupational, lifestyle, or infectious cause**. Oxidative stress is mechanistically central to the *pathophysiology* (see Section 6) but is generated endogenously by the unstable globin rather than by external exposures. Chronic transfusion (a treatment) introduces secondary risks (iron overload; historically transfusion-transmitted HCV/HBV in the broader thalassemia population, e.g. [PMID: 28836463](#)). No infectious agent triggers the disease.

6. Mechanism / Pathophysiology

Ordered causal chain

1. A **heterozygous *HBB* exon-3 missense/nonsense/frameshift mutation** *leads to* production of an mRNA that (for NMD-escaping/elongating alleles) is translated into a **structurally abnormal, hyperunstable β -globin chain** ([PMID: 1971109](#), [PMID: 34271589](#)).
2. The abnormal β -globin **binds heme and precipitates**, forming **protease-resistant aggregates / inclusion (Heinz) bodies** *because* it escapes normal proteolytic clearance ([PMID: 1971109](#)).
3. In parallel, defective β -globin **results in an α/β chain imbalance**, leaving **excess unpaired α -globin**; AHSP normally escorts free α -globin, but the excess **overwhelms this chaperone**, so unpaired α -globin also precipitates ([PMID: 40655320](#), [PMID: 31894534](#)).
4. The **erythroid protein quality-control (PQC) system** — ubiquitin–proteasome (Nrf1-driven) plus compensatory **ULK1-mediated autophagy** — attempts to degrade the aggregates; **dominance reflects the failure/escape of this clearance** ([PMID: 22427201](#)).
5. Precipitated globin **with attached heme accumulates**, *leading to* **reactive oxygen species (ROS) and oxidative membrane damage** in erythroblasts ([PMID: 29180398](#), [PMID: 25724329](#)).
6. Oxidative stress *results in* **apoptosis of erythroid precursors → ineffective erythropoiesis** in the bone marrow ([PMID: 25724329](#)). (Branch: *GDF11/ActRIIA* signaling amplifies this via an ROS-driven autocrine loop involving α -globin precipitation — [PMID: 24658077](#).)

7. Surviving inclusion-bearing red cells that reach circulation are **removed predominantly by the spleen** (cordal-macrophage phagocytosis in the red pulp), *leading to hemolytic anemia and splenomegaly* (PMID: 142356).
8. Chronic anemia and ineffective erythropoiesis *lead to compensatory HbF elevation*, increased intestinal iron absorption, and **secondary iron overload** — the downstream clinical manifestations (thalassemia-intermedia phenotype).

Detail by category

- **Molecular pathways / cellular processes:** protein aggregation, ubiquitin–proteasome degradation, macroautophagy (ULK1), oxidative-stress signaling (Nrf1/Nrf2), Fas/Fas-ligand apoptosis of immature erythroblasts, and GDF11–ActRIIA (TGF- β -superfamily) signaling in ineffective erythropoiesis. β -thalassemia "fits into the broader framework of protein-aggregation disorders that use PQC pathways as cell-protective mechanisms" (PMID: 22427201).
- **Protein dysfunction:** misfolding, heme-bound aggregation, protease resistance (gain-of-toxic-function / dominant-negative).
- **Immune involvement:** not autoimmune; splenic macrophage phagocytosis is the effector of hemolysis.
- **Tissue-damage mechanism:** oxidative stress $\rightarrow$ apoptosis (marrow) and macrophage-mediated erythrophagocytosis (spleen).

Upstream vs downstream: the *HBB* mutation and unstable-globin aggregation are **upstream**; ROS, ineffective erythropoiesis, splenic hemolysis, HbF compensation, and iron overload are **downstream**.

Suggested GO terms: GO:0034976 (response to endoplasmic reticulum stress), GO:0006979 (response to oxidative stress), GO:0043161 (proteasome-mediated ubiquitin-dependent protein catabolic process), GO:0006914 (autophagy), GO:0006915 (apoptotic process), GO:0043249 (erythrocyte maturation), GO:0030218 (erythrocyte differentiation). **Suggested CL terms:** CL:0000765 (erythroblast), CL:0000764 (erythroid progenitor cell), CL:0000232 (erythrocyte), CL:0000235 (macrophage — splenic red-pulp).

7. Anatomical Structures Affected

- **Primary tissue/organ: bone marrow / erythroid lineage** (UBERON:0002371 bone marrow) — site of ineffective erythropoiesis; **blood** (UBERON:0000178).
 - **Spleen** (UBERON:0002106) — primary site of destruction of inclusion-bearing red cells and cause of splenomegaly ([PMID: 142356](#)); the **red pulp** is the specific compartment.
 - **Liver** (UBERON:0002107) — secondary iron deposition; minor Kupffer-cell phagocytosis; extramedullary hematopoiesis in severe cases.
 - **Secondary/systemic:** endocrine glands, heart, and bones (from iron overload and marrow expansion) — as in the broader thalassemia-intermedia spectrum.
 - **Cell populations (CL):** erythroblasts (CL:0000765), erythrocytes (CL:0000232), splenic red-pulp macrophages (CL:0000235).
 - **Subcellular / GO Cellular Component:** cytosol (site of globin aggregation; GO:0005829), proteasome complex (GO:0000502), autophagosome (GO:0005776); Heinz bodies are membrane-associated cytoplasmic inclusions.
 - **Lateralization:** systemic/bilateral (a blood/marrow disorder); splenomegaly involves the single left-sided organ.
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8. Temporal Development

- **Onset:** typically **childhood**, often insidious with chronic hemolytic anemia; some cases recognized in adolescence/adulthood. Reported probands include adolescents ([PMID: 24432801](#)) and adults (a 41-year-old with a novel frameshift, [PMID: 39103314](#)).
 - **Onset pattern: chronic**, congenital in genetic terms; clinically insidious.
 - **Progression / course: chronic, lifelong, generally stable-to-progressive**, with progressive splenomegaly and cumulative iron overload if untreated. Severity is **variable even within families** carrying identical alleles ([PMID: 8184583](#)).
 - **Remission / intervention windows:** no spontaneous remission; **splenectomy** can produce "almost complete recovery from hemolysis" and transfusion independence ([PMID: 142356](#), [PMID: 42261228](#)); curative HSCT/gene therapy offers a one-time intervention.
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9. Inheritance and Population

- **Inheritance: autosomal dominant** — a single *HBB* allele causes disease; **de novo** mutations occur, explaining sporadic/isolated cases.
 - **Penetrance / expressivity**: high penetrance but **variable expressivity**, strongly influenced by α -globin genotype and HbF ([PMID: 8184583](#), [PMID: 28651846](#)).
 - **Anticipation / mosaicism**: not a repeat-expansion disorder (no anticipation); germline mosaicism is theoretically possible but not a defining feature.
 - **Epidemiology: rare and pan-ethnic**. Overall β -thalassemia is among the most common monogenic diseases (~1.5% of the global population are carriers; ~7% of the world population carry a thalassemia gene, with 300,000–400,000 affected births/year) — but these figures pertain to the *recessive* forms ([PMID: 21082937](#), [PMID: 36367309](#)). The **dominant** (inclusion-body) form is a small minority: one survey noted only ~12 families worldwide known with a third-exon nonsense β -thal mutation, and it appears in isolated families/de novo cases across ethnic groups.
 - **Why pan-ethnic (not malaria-linked)**: because heterozygotes are symptomatic, the dominant alleles are **not maintained by malaria selection** — "the absence of selective preference of these mutations in malaria infested areas as a result of serious clinical manifestations in heterozygotes" ([PMID: 8184583](#)). This distinguishes it from recessive β -thalassemia's geographic clustering.
 - **Sex ratio / age distribution**: no strong sex predilection (autosomal); presents across childhood-to-adult ages.
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10. Diagnostics

Laboratory / hematologic. - CBC: microcytic, hypochromic indices with anemia; reticulocytosis; abnormal red-cell morphology (target cells, nucleated RBCs). - **Heinz-body / inclusion-body preparation** (supravital stain, e.g., methyl violet): hallmark of unstable-globin disorders — inclusions may be sparse before splenectomy and abundant after ([PMID: 7022469](#)). - **Heat-instability / isopropanol stability tests**: positive for unstable hemoglobins ([PMID: 24074398](#)). - **Hemoglobin analysis** by HPLC/capillary electrophoresis: elevated **HbA2** and often **HbF**; abnormal peaks for some variants (though many unstable variants are

electrophoretically silent). - **Mass spectrometry** (MALDI-TOF) can quantify globin-chain imbalance as a rapid screen ([PMID: 35098837](#)). - Hemolysis markers: unconjugated hyperbilirubinemia, elevated LDH, low haptoglobin.

Genetic testing (definitive). Single-gene *HBB* sequencing is the diagnostic gold standard and is required to identify the exon-3/frameshift variant, since many causal variants are not detectable by protein methods. **α -globin (*HBA1/HBA2*) genotyping** should accompany it to assess the protective α -thalassemia modifier. Gene panels or WES/WGS can be used when *HBB* Sanger sequencing is non-diagnostic. In-silico protein modeling can support pathogenicity of novel variants ([PMID: 34271589](#)).

Clinical criteria / differential diagnosis. Differentiate from: recessive β -thalassemia intermedia/major; other **congenital Heinz-body hemolytic anemias / unstable hemoglobins** (e.g., Hb Köln $\beta 98\text{Val}\rightarrow\text{Met}$, [PMID: 24074398](#)), G6PD deficiency and other enzymopathies, and hereditary spherocytosis. Family history of a dominantly transmitted hemolytic anemia plus *HBB* sequencing resolves the diagnosis.

Screening. For the dominant form, **cascade family testing** after an index case is the key strategy; classical carrier screening (aimed at recessive β -thal) does not detect dominant alleles in silent carriers because there are none.

11. Outcome / Prognosis

- **Course:** chronic, lifelong hemolytic/dyserythropoietic anemia of **intermediate severity** (thalassemia-intermedia-like). Most affected individuals are **not transfusion-dependent from birth** but may require intermittent or (in severe variants) regular transfusion.
- **Complications:** progressive splenomegaly/hypersplenism, cholelithiasis (chronic hemolysis), secondary **iron overload** with attendant endocrine, hepatic, and cardiac risk (as in NTDT/thalassemia intermedia — [PMID: 36295656](#), [PMID: 28589785](#)), extramedullary hematopoiesis, and thrombosis risk in the broader NTDT spectrum.
- **Recovery / prognostic modifiers:** **splenectomy** can normalize hemolysis and confer transfusion independence in unstable-hemoglobin disease ([PMID: 142356](#), [PMID: 42261228](#)). **Co-inherited α -thalassemia and high HbF** predict milder disease ([PMID: 24432801](#), [PMID: 28651846](#)). Prognosis is generally favorable with modern supportive care, though rare severe unstable-hemoglobin cases have had fatal outcomes from hemosiderosis ([PMID: 6162731](#)).

- **Mortality:** no disease-specific survival statistics exist for this rare subtype; outcomes track the thalassemia-intermedia/NTDT literature and depend on iron-overload management.

12. Treatment

Supportive / foundational care. Red-cell **transfusion** as needed, **iron chelation** (deferroxamine, deferasirox, deferiprone) for iron overload, folate supplementation, and **splenectomy** for hypersplenism/transfusion burden — the latter especially effective in inclusion-body/unstable-hemoglobin disease ([PMID: 142356](#), [PMID: 42261228](#)).

HbF induction. Hydroxyurea and sirolimus (rapamycin) induce γ -globin/HbF; sirolimus co-induces AHSP and ULK1-autophagy in patient erythroid cells, addressing α -globin excess ([PMID: 40655320](#), [PMID: 38731008](#)).

Disease-modifying agents. **Luspatercept** (ActRIIB ligand trap / erythroid maturation agent) and **mitapivat** (pyruvate-kinase activator) are approved and "have demonstrated clinically meaningful improvements in hemoglobin levels and reduction of transfusion burden" ([PMID: 42584024](#), [PMID: 34889443](#)).

Curative / advanced therapeutics. **Allogeneic HSCT** (curative in eligible patients); **gene addition** with **betibeglogene autotemcel (beti-cel)** and **CRISPR-based *BCL11A* editing** with **exagamglogene autotemcel (exa-cel)** — approved and achieving "high rates of durable transfusion independence" ([PMID: 42584024](#), [PMID: 34493145](#)). Note: gene-*addition* corrects globin-chain imbalance but does not remove a dominant toxic allele; allogeneic HSCT and (conceptually) allele-directed editing more fully address a dominant-negative product — an important consideration specific to this subtype.

Suggested NCIT terms (verify codes): Luspatercept, Mitapivat, Hydroxyurea (C577), Deferasirox (C29331), Deferoxamine, Splenectomy (C51772), Hematopoietic Stem Cell Transplantation (C15431), Gene Therapy (C15254), Red Blood Cell Transfusion.

13. Prevention

- **Primary prevention:** as a dominant/often de novo monogenic disease, population carrier screening is less applicable than for recessive β -thal. **Genetic counseling** for affected individuals (50% transmission risk per pregnancy) and **cascade family testing** are central.
 - **Reproductive options: prenatal diagnosis** and **preimplantation genetic testing (PGT)** for the known familial *HBB* variant.
 - **Secondary/tertiary prevention:** early detection of iron overload (ferritin, MRI T2*), timely chelation, vaccination and antibiotic prophylaxis after splenectomy (encapsulated-organism protection), and monitoring for endocrine/cardiac complications.
 - **Public health:** premarital/genetic screening programs (e.g., Saudi Arabia, [PMID: 39073533](#)) target recessive hemoglobinopathies primarily; they do not prevent dominant/de novo cases.
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14. Other Species / Natural Disease

- **Taxonomy / orthologs:** Human *HBB* (NCBI Gene 3043). Orthologous β -globin genes exist across mammals (mouse *Hbb* cluster). β -globin structure and the α/β balance are evolutionarily conserved, which is why murine β -thalassemia models recapitulate core mechanisms.
 - **Natural disease in animals:** unstable-hemoglobin/thalassemia-like syndromes are not a prominent naturally occurring veterinary disease for the dominant human variants specifically; comparative data are limited. No zoonotic potential (non-infectious genetic disease).
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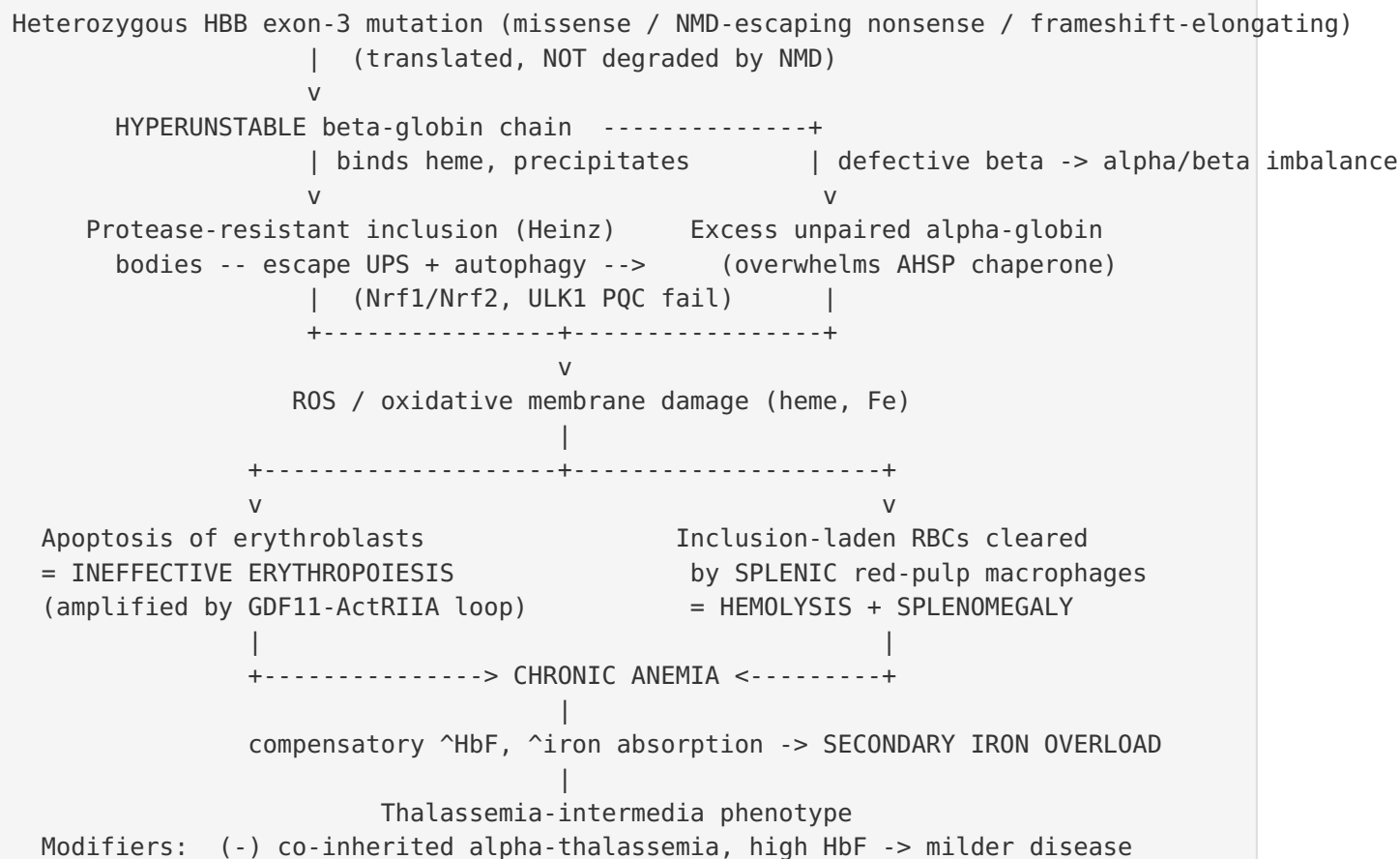
15. Model Organisms

- **Mouse models** are the principal system. β -thalassemia mice reproduce **ineffective erythropoiesis, α -globin precipitation, oxidative stress, and iron dysregulation**, and have been used to validate mechanistic and therapeutic hypotheses: e.g., an ActRIIA ligand trap (RAP-011) correcting ineffective erythropoiesis and GDF11 biology ([PMID: 24658077](#)); heme-oxygenase inhibition ameliorating anemia/iron overload ([PMID: 29180398](#)); and

integrated PQC (proteasome + autophagy) regulation of free α -globin in murine β -thalassemia (PMID: 22427201).

- **In-vitro / cellular models:** K562 and primary erythroid precursor cells (ErPCs) for AHSP/Nrf2/ULK1 studies (PMID: 35092867, PMID: 40655320); HEK-293T transfection to dissect mutant-mRNA/protein stability for specific dominant alleles (PMID: 34271589); patient nucleated erythrocytes for functional characterization.
 - **Model characteristics/limitations:** murine models capture the chain-imbalance/ineffective-erythropoiesis axis well but are usually engineered as loss-of-function/recessive backgrounds; **faithful modeling of a translated toxic dominant-negative β -globin (knock-in of a human hyperunstable allele)** is a gap.
 - **Resources:** MGI (mouse), Cellosaurus (K562), plus patient-derived iPSC/erythroid-differentiation platforms.
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Mechanistic Model / Interpretation



The unifying insight from this investigation is that **dominant β -thalassemia is a proteostasis disorder of the erythron**: dominance is not about quantitative haploinsufficiency but about a **translated, aggregation-prone, protease-resistant globin that escapes protein quality control** and acts as a toxic dominant-negative. This reframes it alongside protein-aggregation diseases and explains (a) why only specific exon-3/frameshift alleles are dominant, (b) why co-inherited α -thalassemia (which lowers the competing α -globin burden) is protective, and (c) why gene-*addition* therapy — which supplies normal β -globin but leaves the toxic allele intact — may be mechanistically less complete than allogeneic HSCT for this subtype.

Evidence Base

PMID	Title (abbrev.)	Contribution
1971109	<i>Molecular basis for dominantly inherited inclusion body beta-thalassemia</i>	Foundational: exon-3 <i>HBB</i> mutations; aggregates resistant to proteolysis
7693620	<i>Hb Hradec Kralove $\beta 115\text{Ala}\rightarrow\text{Asp}$</i>	Dominant phenotype; hematologic picture; splenectomy; $\uparrow$ HbF
8184583	<i>Dominant β-thal alleles in Czech/Slovak population</i>	Nonsense exon-3 alleles; rarity (~12 families); non-malaria rationale; Heinz bodies; variable severity
34957901	<i>Hb Dieppe</i>	Confirms exon-3 missense $\rightarrow$ unstable globin $\rightarrow$ dominant phenotype
34271589	<i>HBB:c.313delA elongated β-globin</i>	Frameshift/elongation; mRNA-stability + protein-degradation determinants
24432801	<i>Hb Grand Junction (HUH)</i>	HUH = dominant β -thal synonym; α -thalassemia co-inheritance is protective
28651846	<i>Molecular basis of β-thal / targets</i>	HbF + α -globin genotype predict severity
22427201	<i>Integrated PQC of free α-globin in murine β-thal</i>	Proteasome + autophagy clearance; protein-aggregation framework
142356	<i>Ultrastructure of spleen/liver in unstable-Hb anemia</i>	Spleen is site of clearance; splenectomy $\rightarrow$ recovery
42261228	<i>Unstable Hb Perth managed with splenectomy</i>	Clinical evidence for splenectomy efficacy
25724329	<i>Role of α-Hb chaperone</i>	ROS $\rightarrow$ apoptosis $\rightarrow$ ineffective erythropoiesis
29180398	<i>Heme-oxygenase inhibition in β-thal mice</i>	Unpaired globin + heme $\rightarrow$ oxidative erythroblast death
24658077	<i>ActRIIA ligand trap corrects ineffective erythropoiesis</i>	GDF11/ROS/ α -globin amplification loop; luspatercept rationale
40655320	<i>ULK1 + AHSP co-induction (sirolimus)</i>	Autophagy/AHSP response to α -globin excess
42584024		

PMID	Title (abbrev.)	Contribution
	2026 Update on Clinical Trials in β -Thalassemia	Approved disease-modifying (luspatercept, mitapivat) + curative gene therapies
21082937	Global burden of β -thal / HbE	1.5% global carrier baseline
36367309	Gujarat screening	7% global thalassemia carriers; 300–400k affected births/yr

Evidence types span **human clinical case/family reports** (variant characterization, phenotype, splenectomy), **mouse models** (mechanism, therapeutics), and **in-vitro cellular systems** (PQC, AHSP/Nrf2/ULK1, mutant mRNA/protein stability).

Limitations and Knowledge Gaps

1. **Rarity → sparse quantitative data.** No robust prevalence/incidence, survival, or QoL statistics exist *specifically* for the dominant subtype; epidemiology is inferred from case series and the broader β -thal/NTDT literature.
2. **Phenotype frequencies are qualitative.** Per-phenotype percentages are not well established for this subtype; HPO frequencies given here are approximate.
3. **Mechanistic evidence is partly extrapolated** from recessive β -thal and general unstable-hemoglobin biology (mouse/in-vitro); direct demonstration of PQC-escape *for each dominant allele* is limited.
4. **Ontology term IDs** for a few entries (e.g., "Heinz bodies," some NCIT drug codes) should be verified by curators against current ontology releases.
5. **Therapeutic subtype-specificity untested.** Whether gene-addition vs. allele-directed editing differs in efficacy for a *dominant toxic* allele is a hypothesis, not a demonstrated clinical result.
6. **No dedicated animal model** faithfully expressing a human hyperunstable dominant β -globin knock-in was identified.

Proposed Follow-up Experiments / Actions

1. **Curate a variant registry** of all reported dominant *HBB* alleles (exon-3 missense/nonsense, frameshift/elongating) with HGVS nomenclature, ClinVar submission, and linked phenotype severity, to enable genotype–phenotype correlation.
 2. **Generate a knock-in mouse or iPSC-erythroid model** expressing a representative hyperunstable human β -globin (e.g., Hb Hradec Kralove) to directly test PQC-escape and dominant-negative toxicity.
 3. **Quantify PQC flux** (proteasome vs. ULK1-autophagy) for individual dominant alleles in patient-derived erythroblasts; test whether pharmacologic autophagy induction (sirolimus) is allele-dependent.
 4. **Systematically test the α -thalassemia modifier** by stratifying reported cases (and any registry) by α -genotype to quantify the protective effect size.
 5. **Evaluate curative-therapy mechanism-fit**: model whether gene-addition adequately dilutes a translated toxic allele vs. approaches that reduce/edit the mutant allele, informing therapy selection for dominant cases.
 6. **Confirm and standardize ontology mappings** (HPO/GO/CL/UBERON/NCIT) for knowledge-base ingestion.
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Report compiled from 9 confirmed findings across 5 iterations and 37 reviewed papers. Evidence prioritizes primary literature with verified abstract quotes; PMIDs are provided for all mechanistic and clinical claims.
